

EBN122 EKSAMEN/EXAMINATION (19 NOVEMBER 2012)**ANTWOORDBLAD/ANSWER SHEET**

Van / Surname:	Memo	Studentennummer: Student number:								
Voorletters / Initials:		Tel. nr. / Tel no.:								
Dissipline / Discipline:		Taal / Language:			Afr	Eng				

Q1	20	Q2	26	Tot:	92 /90
Q3	20	Q4	26		

VRAAG 1 – QUESTION 1 (20)

1.1	$p = 164.5 \text{ mW}$ ✓✓	$W = 10.808 \text{ J}$ ✓✓
1.2	$V = 6.67 \text{ V}$ ✓✓	$W = 16200 \text{ J}$ ✓✓
1.3	$v_1 = 4 \text{ V}$ ✓	$v_2 = 3 \text{ V}$ ✓
	$i_1 = 1 \text{ A}$ ✓	$p = 5 \text{ W}$ ✓
1.4	$R_{in} = 22.11 \Omega$ ✓✓	$v_x = 12 \text{ V}$ ✓✓
	$i_y = 1 \text{ A}$ ✓✓	$i_z = -0.1 \text{ A}$ ✓✓

VRAAG 2 – QUESTION 2 (25)

2.1	$a = 30$ ✓	$b = -25$ ✓
	$c = 35$ ✓	$d = -15$ ✓
2.2	$e = -6$ ✓	$f = -11$ ✓
	$g = -2$ ✓	$h = 5$ ✓
2.3	$V_1 = -1 \text{ V}$ ✓✓	$i_3 = 1 \text{ A}$ ✓✓
2.4	$V_{CE} = 1.932 \text{ V}$ ✓✓	$R_B = 63.7 \text{ k}\Omega$ ✓✓
2.5	$V_{TH} = 2 \text{ V}$ ✓✓	$R_{TH} = 2 \Omega$ ✓✓
2.6	$V_{x(10V)} = -8 \text{ V}$ ✓✓	$V_{x(2A)} = -4 \text{ V}$ ✓✓
2.7	$P_{max} = 911.25 \text{ W}$ ✓✓	

VRAAG 3 – QUESTION 3 (20)

3.1	$a = 7 \checkmark$	$b = -1 \checkmark$
	$c = 10 \checkmark$	$d = 4 \checkmark$
3.2	$\checkmark \checkmark \quad 1k\Omega \leq R_f \leq 2k\Omega \checkmark \checkmark$	
3.3	$a) V_S = -1.5V \checkmark \checkmark$	$b) V_O = -6V \checkmark \checkmark$
3.4	$i(1s) = 0,135 A \checkmark$	$p(1s) = 0.585 W \checkmark$
3.5	$a) i_L = 30mA \checkmark$	$w_L = 4,05\mu J \checkmark$
	$b) v_C = 30V \checkmark$	$w_C = 2,7mJ \checkmark$
3.6	$C_{ab} = 9,75mF \checkmark \checkmark$	

VRAAG 4 – QUESTION 4 (25)

4.1	$a) V_1 \text{ is } \underline{\text{leading}} \quad V_2 \text{ met/with } \underline{170^\circ} \checkmark$	
	$b) V_2 \text{ is } \underline{\text{leading}} \quad V_3 \text{ met/with } \underline{115^\circ} \checkmark$	
4.2	$8,94 \cos(\omega t + 3.43^\circ) \checkmark \checkmark$	
4.3	$a) 2.24 \angle 18.435^\circ \checkmark \checkmark$ (Polêr / Polar)	$b) 3 \checkmark \checkmark$ (Reghoekig / Rectangular)
4.4	$a) i_X(t) = 2 \cos(10t) A \checkmark$	$b) v_X(t) = 4 \cos(10t \pm 180^\circ) \checkmark \checkmark$

4.5 $\omega = 10k\text{rad/s} \checkmark \checkmark$

4.6	$c = -1 - j2 \checkmark$ (Reghoekig / Rectangular)	$d = 2 - j2 \checkmark$ (Reghoekig / Rectangular)
4.7	$e = 8k + j12k \checkmark$ (Reghoekig / Rectangular)	$f = 17k - j13k \checkmark$ (Reghoekig / Rectangular)
	$g = -1000 \checkmark$ (Reghoekig / Rectangular)	$h = 1000 \checkmark$ (Reghoekig / Rectangular)
4.8	$V_{TH} = 8 \angle -135^\circ V \checkmark \checkmark$ (Polêr / Polar)	$Z_{TH} = -4 - j4 \Omega \checkmark \checkmark$ (Reghoekig / Rectangular)